

Haochuan Zhang

Master of Engineering in Computer Science — Oregon State University

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SUMMARY

Applied AI researcher and engineer focused on human mesh recovery, video understanding, and multimodal perception systems.

Founder-level CPO and research lead of a tennis analytics platform that translates state-of-the-art HMR research into a production iOS system with cloud inference.

Strong background in executing large-scale experiments, system integration, and evaluation-driven research iteration.

EDUCATION

Oregon State University

MEng in Computer Science, GPA: 3.97

Corvallis, OR

Sep 2024 – Jun 2026 (expected)

University of Kansas

BS in Computer Science, GPA: 3

Lawrence, KS

Aug 2017 – May 2022

RESEARCH EXPERIENCE

Graduate Research Assistant — Human Mesh & Video Models

Oregon State University, Computer Vision Group

Corvallis, OR

Jan 2025 – Present

- Executed large-scale experiments on monocular 3D human reconstruction using Sam3D-Body, 4D-Humans, and HMR2.0b pipelines.
- Analyzed temporal instability and pose drift in video-based HMR, with quantitative evaluation using PA-MPJPE and joint-wise error breakdowns.
- Implemented and benchmarked temporal aggregation strategies (sliding-window smoothing, recurrent baselines) to improve mesh consistency.
- Prepared the system for future integration of State Space Models (SSM) to model long-range temporal dynamics beyond transformer-style attention.

Research Fellow — Multimodal Robotics Perception (NEST-ARTERD)

University of Tokyo & Japan Atomic Energy Agency (JAEA)

Tokyo & Fukushima, Japan

Mar 2025 – Aug 2025

- Executed perception and mapping experiments combining RGB, depth, LiDAR, and radiation sensing under real-world robotic conditions.
- Validated system robustness and documented reproducible workflows for multimodal sensor integration.

PRODUCT & DEPLOYMENT EXPERIENCE

Chief Product Officer (CPO) & Research Lead — Tennis Human Mesh Platform

Founder-Led Research & Product Initiative

Corvallis, OR

Aug 2025 – Present

- Founded and led a tennis analytics platform bringing research-grade human mesh recovery into a user-facing mobile product.
- Defined the technical roadmap evolving from single-frame HMR toward video-level reconstruction using Sam3D-Body, 4D-Humans, and HMR2.0b.
- Integrated HMR pipelines into an iOS application, enabling users to upload tennis videos and visualize reconstructed 3D meshes.
- Designed and deployed a cloud inference backend on AWS, supporting scalable GPU-based online inference via REST APIs.
- Planned future incorporation of State Space Models (SSM) for long-horizon temporal modeling and motion consistency in sports analysis.

PUBLICATIONS

Y. Xie, J. Shang, Q. Yang, X. Qian, **H. Zhang**, X. Tang,

ARSA-UNet: Atrous residual network based on Structure-Adaptive model for retinal vessel segmentation, Biomedical Signal Processing and Control, Elsevier, 2025.

TECHNICAL SKILLS

Programming: Python (research systems), C++ , Bash

Human Mesh & Vision: Sam3D-Body, 4D-Humans, HMR2.0b, SMPL, video-based reconstruction

Temporal Modeling: recurrent baselines, windowed aggregation, State Space Models (planned integration)

Systems: AWS GPU inference, Docker, Kubernetes, Slurm (HPC)

Evaluation: PA-MPJPE, joint-wise error analysis, ablation studies, reproducible experiment design

RESEARCH INTERESTS

Video-based Human Mesh Recovery and Pose Accuracy

Multimodal Perception and Sports Analytics

Temporal Modeling and World Models